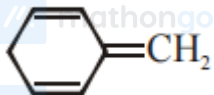


Q1: 16 March (Shift 1) - Single Correct

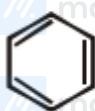
Among the following, the aromatic compounds

are :

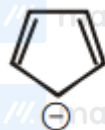
(A)



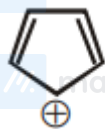
(B)



(C)



(D)



Choose the correct answer from the following options :

(1) (A) and (B) only

(2) (B) and (C) only

(3) (B), (C) and (D) only

(4) (A), (B) and (C) only

Questions with Answer Keys

MathonGo

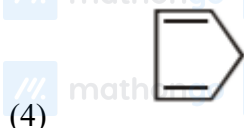
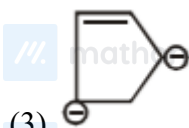
Q2: 16 March (Shift 1) - Single Correct

In chromatography technique, the purification of compound is independent of :

- (1) Mobility or flow of solvent system
- (2) Solubility of the compound
- (3) Length of the column or TLC Plate
- (4) Physical state of the pure compound

Q3: 17 March (Shift 1) - Single Correct

Which of the following is an aromatic compound?



Q4: 17 March (Shift 1) - Single Correct

Mesityl oxide is a common name of :

- (1) 2,4 -Dimethyl pentan- 3 -one
- (2) 3-Methyl cyclohexane carbaldehyde
- (3) 2 -Methyl cyclohexanone
- (4) 4-Methyl pent-3-en-2-one

Q5: 17 March (Shift 2) - Single Correct

Questions with Answer Keys

MathonGo

The correct pair(s) of the ambident nucleophiles is (are):

- (A) AgCN/KCN
- (B) $\text{RCOOAg}/\text{RCOOK}$
- (C) $\text{AgNO}_2/\text{KNO}_2$
- (D) AgI/KI

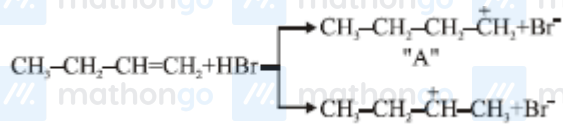
- (1) (B) and (C) only
- (2) (A) only

- (3) (A) and (C) only

- (4) (B) only

Q6: 17 March (Shift 2) - Single Correct

Choose the correct statement regarding the formation of carbocations A and B given :-



- (1) Carbocation B is more stable and formed relatively at faster rate
- (2) Carbocation A is more stable and formed relatively at slow rate
- (3) Carbocation B is more stable and formed relatively at slow rate
- (4) Carbocation A is more stable and formed relatively at faster rate

Q7: 18 March (Shift 1) - Single Correct

Match List-I with List-II :

List-I

(Chemicals)

(a) Alcoholic potassium hydroxide

(b) Pd/ BaSO₄

(c) BHC (Benzene hexachloride)

(d) Polyacetylene

List-II

(Use / Preparation / Constituent)

(i) Electrodes in batteries

(ii) Obtained by addition reaction

(iii) Used for β - elimination reaction

(iv) Lindlar's catalyst

Choose the most appropriate match :

(1) a - ii, b - i, c - iv, d - iii

(2) a - iii, b - iv, c - ii, d - i

(3) a - iii, b - i, c - iv, d - ii

(4) a - ii, b - iv, c - i, d - iii

Q8: 18 March (Shift 1) - Single Correct

Compound with molecular formula C₃H₆O can show :

(1) Positional isomerism

(2) Both positional isomerism and metamerism

(3) Metamerism

(4) Functional group isomerism

Q9: 18 March (Shift 2) - Single Correct

Questions with Answer Keys

MathonGo

Given below are two statements :

Statement I : $\text{C}_2\text{H}_5\text{OH}$ and AgCN both can generate nucleophile.

Statement II : KCN and AgCN both will generate nitrile nucleophile with all reaction conditions.

(1) Statement I is true but statement II is false

(2) Both statement I and statement II are true

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are false

Answer Key

Q1 (2)

Q2 (4)

Q3 (1)

Q4 (4)

Q5 (3)

Q6 (1)

Q7 (2)

Q8 (4)

Q9 (1)

Hints and Solutions

MathonGo

Q1 (2)

(A) Non-Aromatic

(B) Aromatic

(C) Aromatic

(D) Anti-Aromatic

Q2 (4)

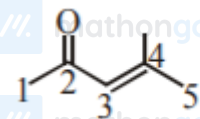
In chromatography technique, the purification of a compound is independent of the physical state of the pure compound.

Q3 (1)



→ Aromatic compound

Q4 (4)



IUPAC [4-Methylpent-3-en-2-one]

Q5 (3)

Ambident nucleophile

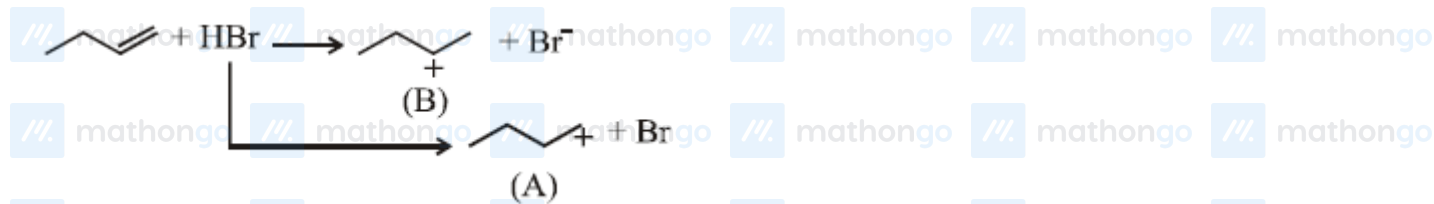
(A) KCN & AgCN

(C) AgNO₃ & KNO₂

Q6 (1)

Hints and Solutions

MathonGo



This is more stable due to secondary cation formation and formed with faster rate due to low activation energy.

Q7 (2)

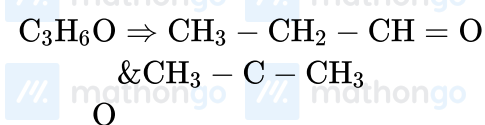
(a) Alcoholic potassium hydroxide $\rightarrow$ used for β -elimination

(b) Pd/BaSO₄ $\rightarrow$ Lindlar's catalyst

(c) BHC (Benzene hexachloride) $\rightarrow$ Obtained by addition reactions

(d) Polyacetylene $\rightarrow$ Electrodes in batteries

Q8 (4)



They are functional group isomerism.

Q9 (1)